Meta-SWAG: Bayesian Posterior Approximation over Markovian Learning Trajectories for Multi-Agent Policy Gradient and LLM Alignment

Abstract

Posterior approximation is now a standard tool for uncertainty in supervised deep learning, but comparable machinery is still missing in settings where the training trajectory is generated by an evolving learning process rather than a fixed objective. We introduce **Meta-SWAG**, a SWAG-style Gaussian posterior approximation over checkpoints drawn from a **Markovian learning trajectory**. The key observation is that two settings that are usually treated separately, multi-agent policy gradient and LLM alignment, both produce adaptive checkpoint sequences whose effective objective shifts over time. This lets us define a coherent posterior $q_T(\theta) = \mathcal{N}(\mu_T, \Sigma_T)$ even when raw training is non-stationary. We present two formal results and one conceptual limitation result: under explicit evidence-variance alignment assumptions, Bayesian model averaging improves over point estimates through a variance reduction term; the posterior covariance tracks local trajectory geometry near stable regions; and posterior quality is bottlenecked by a self-knowledge gap between an updater and the policy model it relies on. We also introduce Goodhart-resilient weighting rules that prevent posterior collapse onto proxy-maximizing checkpoints. Empirically, the current draft pairs completed matrix and iterated-game scaffolds with a LoRA-based DPO/PPO/GRPO protocol that remains partially pending.

1 Introduction

Uncertainty estimates for learned policies remain underdeveloped in exactly the domains where they matter most. In multi-agent reinforcement learning, learning dynamics are shaped by the simultaneous adaptation of other agents [1, 3, 4]; in LLM alignment, the object being optimized is a changing policy produced by DPO, PPO, or GRPO updates rather than a fixed supervised predictor [6, 8, 9]. Both settings offer local convergence or optimization stories, but neither comes with a standard posterior approximation that can be used for model averaging, geometry diagnostics, or uncertainty-aware deployment.

SWAG and related subspace methods are attractive candidates because they give cheap Gaussian posteriors from training trajectories [5]. The obstacle is that vanilla SWAG presumes a stationary loss landscape and approximately exchangeable late iterates. Those assumptions fail in raw MARL and raw fine-tuning trajectories because the effective objective changes as the environment, the peer policies, or the aligned policy itself changes. Our central claim is that the failure is not fatal: a SWAG-style posterior can be adapted to **Markovian learning trajectories** where checkpoints evolve under a fixed update operator and a shifting effective objective. The LLM case is therefore not a claim that the optimizer is an agent; it is the milder claim that fine-tuning induces a checkpoint process whose states evolve Markovianly under repeated alignment updates.

This paper introduces Meta-SWAG, a Gaussian posterior approximation over adaptive checkpoint sequences. The method is built around one clean object, $q_T(\theta) = \mathcal{N}(\mu_T, \Sigma_T)$, with the posterior restricted to LoRA adapter parameters in the LLM setting. We make three claims. First, under an explicit evidence-variance alignment assumption, Bayesian model averaging over the posterior reduces behavioral variance relative to point estimates. Second, the posterior covariance captures local trajectory geometry, so uncertainty structure becomes a diagnostic rather than just an interval estimate. Third, posterior quality is constrained by the quality of the updater's internal policy model; in the alignment setting this provides a conceptual bound on how far posterior averaging can compensate for misspecified self-models.

Empirically, we design the paper around two validations. The first uses matrix games and Kim-style iterated games to test the variance and geometry claims in a controlled setting. The second targets LoRA-based alignment of a Llama-family model under DPO, PPO, and GRPO, with reward-overoptimization measured by

a gold reward model and best-of-n sampling [2, 7]. The current draft keeps the full LLM protocol and figure structure even where final runs are still pending, so the contribution should be read as a sharpened method paper with completed small-scale evidence rather than as a fully finished large-scale empirical study.

2 Background

2.1 Markovian Learning Trajectories in MARL and LLM Alignment

Let θ_t denote the checkpoint after t learning updates and let $\mathcal{T}$ denote the update operator that maps one checkpoint to the next. The trajectory state is simply θ_t, and the transition is the optimizer-induced map $\theta_{t+1} = \mathcal{T}(\theta_t, \xi_t)$ under minibatch and environment noise ξ_t. What changes over time is not the definition of the update rule, but the effective objective seen by the checkpoint sequence.

In multi-agent policy gradient, the checkpoint θ_t contains the joint policy parameters of all agents, and the effective objective shifts because peer adaptation changes the learning environment [3, 4]. In LLM alignment, θ_t is the current aligned model checkpoint, and the effective objective shifts because repeated policy updates change the policy whose outputs define the next gradient step [6, 8, 9]. This shared sequential structure is what motivates applying the same posterior construction across both domains.

The importance of this trajectory view is conceptual and technical. Conceptually, it says that what we should be uncertain about is not a single static predictor but a trajectory of adaptive policies. Technically, it restores the object over which posterior approximation is coherent: the distribution of checkpoints induced by a learning process with Markovian dependence across updates.

2.2 SWAG Beyond Stationary Supervised Losses

Standard SWAG approximates the posterior over parameters by fitting a Gaussian to late training iterates. The approximation is compelling when the iterates come from a relatively stationary neighborhood of a single basin, so that first and second moments summarize the local geometry well. In raw RL or raw fine-tuning, however, these assumptions fail for two reasons. First, the effective objective is non-stationary because the data distribution changes with the policy. Second, late iterates need not be exchangeable when the update rule is itself part of the object being studied.

The Markovian trajectory framing resolves both failures at the correct level. The trajectory we summarize is no longer a raw sequence of arbitrary task-level gradients; it is a dependent sequence of checkpoints generated by a single update dynamic. Meta-SWAG therefore approximates uncertainty over adaptive policies rather than over static predictors. This is the main difference between our use of SWAG and the standard supervised setting: we keep the cheap diagonal-plus-low-rank approximation, but we change the object whose moments are being estimated.

2.3 Bayesian Alignment Landscape

Recent work has brought Bayesian ideas into LLMs through local curvature approximations, adapter-space posteriors, and Bayesian reward models [10, 11]. Meta-SWAG complements rather than replaces these approaches. Laplace-LoRA approximates posterior uncertainty around one adapted solution using local curvature in LoRA space; Bayesian reward-model approaches place uncertainty on the learned reward function; Meta-SWAG instead places uncertainty on the aligned policy trajectory itself.

Method	Posterior object	Extra models	Trajectory-aware	DPO/PPO/GRP
Laplace-LoRA	Local adapter posterior around one solution	No	No	Usually yes
Bayesian reward modeling	Reward model posterior	Yes	Indirectly	Yes
Meta-SWAG	**Posterior over aligned policy checkpoints**	**No**	**Yes**	**Yes**

Table 1: Positioning Meta-SWAG against nearby Bayesian alignment methods.

The distinction matters because reward uncertainty and policy uncertainty answer different questions. Reward-model posteriors quantify uncertainty about the evaluator. Meta-SWAG quantifies uncertainty about the policy that will actually be sampled at deployment time.

3 Meta-SWAG

3.1 Posterior Definition

Let $\{\theta_t\}_{t=1}^{T}$ be checkpoints sampled from a Markovian learning trajectory after burn-in, and let m_t be a scalar evidence score for checkpoint t. In MARL, m_t is a held-out meta-value estimate; in LLM alignment, m_t

is a held-out alignment objective such as validation preference accuracy or reward-model score. A naive softmax weighting

$$w_t = \frac{\exp(\beta m_t)}{\sum_{s=1}^{T} \exp(\beta m_s)}, \tag{1}$$

can collapse onto a single proxy-maximizing checkpoint. To prevent this Goodhart-style posterior collapse, Meta-SWAG uses a Goodhart-resilient weighting rule. Our preferred default is **ESS-constrained softmax weighting**: choose β adaptively so that the effective sample size

$$ESS(w) = \frac{(\sum_t w_t)^2}{\sum_t w_t^2} \tag{2}$$

stays above a fixed fraction of the retained checkpoints. As a simpler ablation, we also consider a satisficing threshold rule that assigns uniform mass to all checkpoints above a validation threshold.

Meta-SWAG defines a Gaussian posterior

$$q_T(\theta) = \mathcal{N}(\mu_T, \Sigma_T), \tag{3}$$

with evidence-weighted mean

$$\mu_T = \sum_{t=1}^{T} w_t \theta_t, \tag{4}$$

second-moment accumulator

$$v_T = \sum_{t=1}^{T} w_t (\theta_t - \mu_T) \odot (\theta_t - \mu_T), \tag{5}$$

and low-rank deviation matrix $D_T = [\sqrt{w_t}(\theta_t - \mu_T)]_{t \in \mathcal{K}}$ built from the last $|\mathcal{K}|$ checkpoints retained for the subspace estimate. We then use the diagonal-plus-low-rank covariance

$$\Sigma_T = \frac{1}{2}\mathrm{diag}(v_T) + \frac{1}{2(|\mathcal{K}|-1)} D_T D_T^\top. \tag{6}$$

For LLM alignment, θ_t is restricted to LoRA adapter parameters. This keeps the posterior in the subspace actually modified by alignment and avoids pretending to infer uncertainty over the frozen base model weights.

Figure 1 (placeholder): Domain Bridge for Meta-SWAG.

Left: multi-agent policy gradient as a Markovian learning trajectory over joint policy checkpoints.
Center: both domains feed the same Goodhart-aware Gaussian posterior $q_T(\theta)$.
Right: LLM alignment as a Markovian learning trajectory over LoRA checkpoints, where updates are DPO/PPO/GRPO steps.

Figure 1: The paper's unification claim: the same posterior construction applies to two adaptive Markovian checkpoint processes.

3.2 Algorithm

Algorithm 1: Meta-SWAG

1. Inputs: checkpoint stream $\{\theta_t\}_{t=1}^T$, evidence scores $\{m_t\}_{t=1}^T$, weighting rule, retained subspace index set $\mathcal{K}$, sample count S.
2. Compute weights using either ESS-constrained softmax weighting or thresholded satisficing weights.
3. Accumulate weighted mean $\mu_T = \sum_t w_t \theta_t$.
4. Accumulate diagonal moment $v_T = \sum_t w_t (\theta_t - \mu_T) \odot (\theta_t - \mu_T)$.
5. Form low-rank deviation matrix D_T from retained weighted deviations.
6. Set $\Sigma_T = \frac{1}{2}\mathrm{diag}(v_T) + \frac{1}{2(|\mathcal{K}|-1)} D_T D_T^\top$.
7. Draw posterior samples $\theta^{(s)} \sim \mathcal{N}(\mu_T, \Sigma_T)$ for $s = 1, \ldots, S$.
8. Average policy outputs or downstream scores across posterior samples.

Outputs: posterior $q_T(\theta) = \mathcal{N}(\mu_T, \Sigma_T)$, posterior samples, Bayesian model average.

The training-time cost is the same order as SWAG: one running mean, one diagonal accumulator, and a bounded low-rank buffer. The inference-time cost is S forward passes or policy evaluations. In the LLM setting, the dominant practical choice is to sample LoRA adapters and merge them into the frozen base model at evaluation time.

3.3 Design Choices

Why evidence-weight the mean? A plain average treats all checkpoints as equally informative, which is undesirable when the meta-objective clearly prefers some stages of training. Evidence weighting makes the posterior predictive closer to the checkpoints that score well on held-out meta-value or alignment criteria

without collapsing fully to a point estimate.

Why guard against Goodhart collapse? If m_t is itself a proxy objective, naive weighting can over-concentrate posterior mass on the same over-optimized checkpoint that a point estimate would select. ESS-constrained weighting makes this failure mode explicit and prevents collapse by guaranteeing posterior support over multiple retained checkpoints. Thresholded satisficing provides a simpler alternative that preserves variation across all "good enough" models.

Why keep a low-rank covariance term? The diagonal term captures marginal uncertainty, but the low-rank factor is what carries basin geometry. The second theorem will use this structure directly: if the trajectory stays in a stable basin, the dominant eigenvectors of Σ_T track the locally soft and stiff directions of the meta-objective.

Why a joint posterior rather than per-agent posteriors? In MARL the relevant object is the joint policy, not a bag of isolated marginals. In alignment the same logic reappears as base-policy and aligner coupling: what matters is uncertainty over the adapted policy that results from the update process, not uncertainty over individual training steps in isolation.

4 Theory

4.1 Variance Reduction Through Bayesian Model Averaging

Assumption: Evidence-variance alignment. For the retained checkpoints, suppose the weighting rule is monotone in an evidence score m_t and that higher evidence corresponds to lower checkpoint-level predictive variance, i.e., $m_t \geq m_s \Rightarrow V_t \leq V_s$. This is an idealized assumption, not a generic property of arbitrary validation metrics.

Theorem 1 (Idealized variance reduction under evidence-variance alignment)**.** *Let q_T be the Meta-SWAG posterior constructed from a Markovian learning trajectory whose checkpoints satisfy the regularity conditions used for evidence-weighted policy gradient, and let $\bar{\pi}_{q_T}$ denote the posterior predictive policy. Under evidence-variance alignment, the within-checkpoint component of the predictive variance of $\bar{\pi}_{q_T}$ is bounded by the corresponding uniformly weighted checkpoint average up to the harmonic-to-arithmetic mean ratio of checkpoint variances:*

$$\mathrm{Var}_{\mathrm{within}}(\bar{\pi}_{q_T}) \leq \frac{\mathrm{HM}(V_1, \ldots, V_T)}{\mathrm{AM}(V_1, \ldots, V_T)} \mathrm{Var}_{\mathrm{within}}(\pi_{\mathrm{unif}}) \leq \mathrm{Var}_{\mathrm{within}}(\pi_{\mathrm{unif}}), \tag{7}$$

where V_t denotes checkpoint-level predictive variance and π_{unif} denotes the uniformly weighted checkpoint ensemble. If the between-checkpoint term remains controlled inside one stable region, this also yields a reduction relative to nearby point estimates.

Proof sketch. The predictive variance of the Bayesian model average decomposes into a within-checkpoint term and a between-checkpoint term. Under evidence-variance alignment, monotone evidence weighting assigns larger mass to lower-variance checkpoints, so the within-checkpoint term is controlled by the harmonic mean rather than the arithmetic mean. The argument mirrors the AM-HM improvement from evidence-weighted policy gradients, but the random object is now the posterior predictive policy rather than the gradient estimator. Extending the comparison from a uniformly weighted ensemble to a single point estimate requires the additional single-region control stated above.

Corollary 1 (Alignment)**.** *In LoRA-based LLM alignment, Bayesian model averaging over q_T reduces behavioral variance relative to the MAP checkpoint and therefore attenuates reward-overoptimization in best-of-n selection whenever the proxy reward differs across posterior samples.*

Significance. The posterior is useful even before geometry or epistemic diagnostics enter the story: averaging over plausible aligned policies is a cheap defense against brittle point-estimate behavior.

4.2 Posterior Covariance as Basin Geometry

Theorem 2 (Posterior covariance identifies local trajectory geometry)**.** *Suppose the retained trajectory remains in a stable local region of the adaptive optimization process, the dynamics admit a local linearization in that region, and the retained checkpoint process is geometrically mixing. Then, as $T \to \infty$, the Meta-SWAG covariance Σ_T converges to a matrix whose leading eigenspaces agree with the dominant weakly-curved directions of the local trajectory geometry, while concentration in sharper directions scales with the local second-order structure of the effective objective.*

Proof sketch. Near a stable local region, the adaptive dynamics admit a local linearization whose covariance is governed by the Jacobian and Hessian of the

effective objective along the trajectory. The diagonal-plus-low-rank estimator is consistent for the dominant second-moment structure of this local process. Geometric mixing ensures that the checkpoint buffer estimates local covariance rather than transient drift between unrelated regions. The leading eigenvectors therefore align with soft directions of the local orbit or basin, while narrow directions shrink under repeated averaging.

Corollary 2 (Alignment)**.** *In aligned LLMs, eigenvalue concentration in Σ_T can be used as a diagnostic for fragile alignment: broad, symmetric uncertainty suggests a robust basin, whereas a spiked spectrum indicates dependence on a narrow set of adapter directions that may fail under distribution shift.*

Significance. Meta-SWAG turns posterior covariance into an interpretable object: it is not just uncertainty mass, but a compressed summary of the geometry of the learned basin.

4.3 Conceptual Limit: Self-Knowledge Bottlenecks Posterior Quality

The next result is intentionally more interpretive than the first two. It does not characterize the implemented weighting rule directly; instead it formalizes a limitation principle for posterior predictors built from an explicit self-model $\tilde{\pi}$ of the policy being updated.

Theorem 3 (Conceptual self-knowledge bound under an explicit self-model)**.** *Let π_θ be the true policy at checkpoint θ, let $\tilde{\pi}_\theta$ be the optimizer's self-model of that policy, and let $\hat{\pi}_{q_T} = \mathbb{E}_{\theta' \sim q_T}[\tilde{\pi}_{\theta'}]$ be the policy induced by the Meta-SWAG posterior predictive. If*

$$C := \sup_x \left| \log \frac{\tilde{\pi}_\theta(x)}{\hat{\pi}_{q_T}(x)} \right| < \infty, \tag{8}$$

then

$$D_{\mathrm{KL}}(\pi_\theta \,\|\, \hat{\pi}_{q_T}) \leq \left(1 + \frac{C^2}{2}\right) L_{\mathrm{self}}(\theta) + O(T^{-1}), \tag{9}$$

where $L_{\mathrm{self}}(\theta) := D_{\mathrm{KL}}(\pi_\theta \,\|\, \tilde{\pi}_\theta)$ is the self-knowledge gap.

Proof sketch. Decompose $D_{\mathrm{KL}}(\pi_\theta \,\|\, \hat{\pi}_{q_T})$ into $D_{\mathrm{KL}}(\pi_\theta \,\|\, \tilde{\pi}_\theta)$ plus a cross term involving $\log(\tilde{\pi}_\theta / \hat{\pi}_{q_T})$. Bound the cross term by a change-of-measure argument and Pinsker's inequality, then use Young's inequality to isolate the residual into a multiplicative constant on L_{self}. The remaining $O(T^{-1})$ term is the usual finite-sample error from the SWAG moment approximation along a mixing Markovian checkpoint trajectory.

Corollary 3 (Alignment)**.** *Let π be the base LLM policy, $\tilde{\pi}$ the aligner's internal model of the base policy, and $\hat{\pi}_{q_T}$ the aligned posterior predictive policy. Then alignment quality, measured as $D_{\mathrm{KL}}(\pi \,\|\, \hat{\pi}_{q_T})$, is upper-bounded by the aligner's self-knowledge of the base model up to the constant above. Read operationally, this means posterior averaging cannot compensate arbitrarily for a poor internal model of the policy being aligned.*

Significance. The theorem identifies a limitation principle for adaptive systems: better uncertainty estimates cannot outrun a poor internal model of the policy being updated.

5 Experiments

5.1 Matrix-Game Validation

The first empirical block validates the variance and geometry claims in controlled environments where equilibria and basin structure can be inspected directly. Our executable scaffold currently focuses on matching pennies, a 3-action zero-sum variant, and a first Kim-style iterated IPD/RPS adaptation runner with bundled personas. The comparison set is: point estimate, vanilla SWAG over checkpoints, naive softmax Meta-SWAG, and Goodhart-resilient Meta-SWAG with ESS-constrained or thresholded weighting.

For Theorem 1, the main metric is effective predictive variance of the Bayesian model average relative to both a point estimate and a uniformly weighted checkpoint ensemble. The target plot places the empirical variance ratio against the theoretical HM/AM prediction across different levels of checkpoint heterogeneity while also checking whether Goodhart-resilient weighting avoids collapse to single-checkpoint behavior. For Theorem 2, we compute the leading eigenvalues of Σ_T and compare them to analytically or numerically estimated basin width around the convergent region. We also log the top five eigenvalues of the low-rank deviation matrix across runs to test whether the retained trajectory occupies a narrow or broad subspace.

Figure 2 (placeholder): Matrix-game validation.
Left panel: empirical variance ratio versus the predicted HM/AM ratio for point estimate, uniformly weighted SWAG, and Meta-SWAG variants.
Right panel: posterior top-eigenvalue ratio versus basin-width estimate across games and initialization regimes.

Figure 2: Main matrix-game figure for validating Theorems 1 and 2.

5.2 LLM Alignment Validation

The second empirical block tests Meta-SWAG on LoRA-based alignment of a Llama-family model. The draft protocol uses DPO as the headline setting and treats PPO and GRPO as algorithm-agnostic extensions rather than separate experiments [6, 8, 9]. Checkpoints are collected along the fine-tuning trajectory; evidence scores are computed from held-out alignment metrics; the posterior is fit in LoRA space only. The main comparison is between the MAP checkpoint, standard softmax-weighted Meta-SWAG, and Goodhart-resilient Meta-SWAG under ESS-constrained or thresholded weighting.

The first target measurement is reward overoptimization. Following the best-of-n evaluation style used in the alignment literature [2, 7], we compare the gap between a proxy reward and a gold reward model as n increases. The weighting metric, model-selection metric, and final evaluation metric should be reported separately so the Goodhart-mitigation claim is not tautological. The core three conditions are: MAP checkpoint, standard softmax-weighted Meta-SWAG, and Goodhart-resilient Meta-SWAG. A Bayesian reward or Laplace-style policy baseline is included where available [10, 11]. The second target measurement is algorithm-agnosticism: the same posterior construction should improve or stabilize outcomes across DPO, PPO, and GRPO without retraining ensembles for each aligner.

This section is intentionally placeholder-ready. The figure slots, metrics, and evaluation protocol are fixed even when some final runs are pending. Where results are not yet complete, the paper should say so explicitly rather than implying completed experiments.

Figure 3 (placeholder; some rows pending): LLM alignment validation.

Top row: reward-overoptimization gap versus best-of-n for MAP, standard softmax Meta-SWAG, and Goodhart-resilient Meta-SWAG under DPO.
Bottom rows: algorithm-agnostic improvement under DPO, PPO, and GRPO measured by proxy-gold gap and posterior predictive variance.

Rows without completed runs should be explicitly marked **RESULT PENDING**.

Figure 3: Main LLM figure scaffold. The draft keeps this section in the body even if some numeric entries are still pending.

Current draft status. The matrix and iterated-game sections form the completed empirical base of the current draft. The LLM subsection should remain in the body with full protocol and caption text even if the numeric cells are temporarily empty.

6 Related Work

Meta-SWAG sits at the intersection of three threads. The first is posterior approximation in deep learning, especially SWAG, subspace inference, and adapter-space Bayesian methods [5, 11]. Our contribution is not a new low-rank posterior family; it is the claim that a SWAG-style posterior becomes coherent again once the trajectory is treated as a non-stationary Markovian checkpoint process.

The second thread is meta-learning in multi-agent reinforcement learning. Kim-style analyses, LOLA-style opponent shaping, and related methods describe the dynamics of adaptation, but they do not typically expose a cheap posterior over checkpoints that can be sampled at deployment time [1, 3, 4]. Meta-SWAG is intended as a posterior layer that can sit on top of adaptive optimization trajectories, provided checkpoint retention and held-out evidence are available.

The third thread is Bayesian alignment. Reward-model posteriors and Laplace-style approximations quantify uncertainty over evaluators or local parameter neighborhoods [10, 11]. Meta-SWAG instead quantifies uncertainty over the aligned policy trajectory and is therefore closest in spirit to policy-side Bayesian model averaging.

7 Discussion and Limitations

Meta-SWAG is deliberately modest about what it approximates. The posterior is Gaussian, which is unlikely to be adequate when alignment basins are genuinely bimodal or when policy trajectories move between distant equilibria. The covariance interpretation is strongest near convergence, where the local second-order story is meaningful; earlier in training, the low-rank factor is better interpreted as trajectory structure than as calibrated epistemic uncertainty.

The self-knowledge result is also conservative and should be read as conceptual unless an explicit self-model is implemented in the empirical stack. The multiplicative constant $1 + C^2/2$ is easy to derive and easy to explain, but it is unlikely to be tight for expressive policies. Richer divergences or chi-squared style refinements may sharpen the bound, but the current form already isolates the main conceptual point: posterior quality in adaptive systems is bottlenecked by self-knowledge, not just by moment estimation.

The most natural extension is to richer posterior fam-

ilies such as MultiSWAG or mixture approximations over multiple alignment basins. A second direction is to combine Meta-SWAG with causal or mechanistic diagnostics so that posterior samples are not only uncertain but also interpretable. For the present paper, however, the goal is narrower: establish that posterior approximation over non-stationary Markovian learning trajectories is coherent, cheap, and useful in both MARL and LLM alignment.

A Notation and Assumptions

We use θ_t for checkpoints in both application domains.

- In MARL, θ_t denotes the joint policy parameters at meta-time t.
- In LLM alignment, θ_t denotes the LoRA adapter parameters at step t; the base model remains frozen.
- m_t is an evidence score on a held-out meta-objective.
- w_t is the normalized evidence weight derived from m_t.
- $q_T(\theta)$ is the Meta-SWAG posterior after T retained checkpoints.

The draft assumes: (i) Markovian trajectory coherence; (ii) late-trajectory concentration in a region where moment estimation is meaningful; (iii) bounded moments; and (iv) availability of a held-out evidence score for each retained checkpoint.

B Posterior Construction Details

B.1 Evidence Weighting

The draft main text treats naive softmax weighting as a baseline

$$w_t = \frac{\exp(\beta m_t)}{\sum_{s=1}^{T} \exp(\beta m_s)}. \tag{10}$$

This rule is scale-free once m_t values are centered, but by itself it is vulnerable to Goodhart-style posterior collapse when m_t is a proxy objective. The current implementation therefore supports two Goodhart-resilient alternatives: ESS-constrained softmax weighting, which adapts β to preserve a minimum effective sample size, and thresholded satisficing weights, which assign uniform mass to all checkpoints above a validation threshold.

B.2 Diagonal-Plus-Low-Rank Covariance

The covariance estimator is

$$\Sigma_T = \frac{1}{2}\mathrm{diag}(v_T) + \frac{1}{2(|\mathcal{K}|-1)} D_T D_T^\top. \tag{11}$$

The diagonal term controls per-parameter uncertainty; the low-rank term preserves the dominant trajectory subspace. In the LLM setting the computation is performed only in LoRA space, which is critical both computationally and conceptually.

B.3 Posterior Prediction

Given samples $\theta^{(s)} \sim q_T$, posterior prediction uses

$$p(y \mid x, \mathcal{D}) \approx \frac{1}{S} \sum_{s=1}^{S} p(y \mid x, \theta^{(s)}). \tag{12}$$

For MARL, the same averaging is performed over policy outputs or rollout-level statistics. For LLMs, the default deployment object is a Bayesian model average over LoRA-sampled policies rather than a single merged checkpoint.

C Expanded Theory Notes

C.1 Theorem 1 Notes

The first theorem should be read as a posterior analogue of the variance reduction seen in evidence-weighted policy gradients. A longer proof path for the paper version is: decompose predictive variance into within-checkpoint and between-checkpoint components; show that evidence weighting controls the within-checkpoint component by the harmonic mean of checkpoint-level variances; and show that the between-checkpoint term vanishes in the degenerate point-estimate limit and remains controlled whenever the retained trajectory lies in a single basin.

C.2 Theorem 2 Notes

The geometry theorem depends on a local stationary approximation. In the final write-up it will help to isolate two claims that are currently bundled together: the retained trajectory mixes within a stable basin, and the empirical covariance estimates the local second-order geometry of that basin. The theorem does not claim global posterior calibration.

C.3 Theorem 3 Notes

The self-knowledge theorem is the conceptual centerpiece of the paper, so the appendix should carry a slightly more explicit derivation. Start from

$$D_{\mathrm{KL}}(\pi_\theta \,\|\, \hat{\pi}_{q_T}) = D_{\mathrm{KL}}(\pi_\theta \,\|\, \tilde{\pi}_\theta) + \mathbb{E}_{\pi_\theta}\left[\log \frac{\tilde{\pi}_\theta}{\hat{\pi}_{q_T}}\right]. \tag{13}$$

Then identify the first term with $L_{\mathrm{self}}(\theta)$, bound the second term using the uniform log-density ratio constant C, move from total variation to KL using Pinsker, absorb the mixed term with Young's inequality, and

append the finite-sample posterior approximation term $O(T^{-1})$.

D Experiment Protocols

D.1 Matrix Games

Planned environments are matching pennies, a 3-action zero-sum variant, and Kim-style iterated IPD/RPS follow-ups. Planned comparisons are point estimate, vanilla SWAG over retained checkpoints, standard softmax Meta-SWAG, and Goodhart-resilient Meta-SWAG with ESS-constrained or thresholded weighting. Primary metrics are effective predictive variance, HM/AM-predicted versus empirical variance ratio, effective sample size, top-eigenvalue concentration of Σ_T, and a basin-width proxy from convergence under perturbed initialization.

D.2 LLM Alignment

Planned setup: a Llama-family base model, LoRA-only adaptation, DPO as the primary aligner with PPO and GRPO extensions, an Alpaca-style preference benchmark or equivalent, and evaluation through proxy reward, gold reward model, best-of-n gap, posterior predictive variance, effective sample size, and max-weight collapse diagnostics. If results are not finished, the paper should still report the checkpoint collection rule, the evidence score definition, the evaluation metrics, the intended figure captions, and which rows are pending.

E LoRA Tracking Implementation Notes

The implementation should track only the trainable LoRA tensors during alignment. A practical first pass is: save LoRA adapter weights every fixed number of optimizer steps after burn-in; score each retained checkpoint on a held-out alignment split; compute evidence weights from the held-out score; accumulate the weighted mean and diagonal moments online; retain only the latest K centered deviations for the low-rank factor; and sample LoRA adapters at evaluation time before merging them into the frozen base model.

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